

overcome the costs they impose on investors. Investors, on average and over time, will do better with a low-cost index fund.”

PRIVATE EQUITY

Private equity encompasses several long-term illiquid investment strategies. It began as a rebranding of leveraged buyouts after the 1980s. Private equity also includes venture capital, private growth capital, and distressed capital/special situations. The amount of private equity assets under management in 2012 was around \$2 trillion, comparable to the amount in hedge funds.

Buyout funds have been much riskier than the S&P 500 and have historically had significantly higher excess returns. However, according to Higson and Stucke (2012), there has been a downward trend in buyout fund absolute returns over the 28 years ending in 2008. The excess returns of buyout funds have been driven largely by the performance of the top 10% of these funds. It would have been difficult to know ahead of time which ones these were going to be.

Over the past 40 years, venture capital funds have had annual gains of 13.4%, versus 12.4% for the S&P 500 Index and 14.4% for the S&P SmallCap Growth Index. Venture capital funds have had higher volatility, illiquidity, and survivorship risk. Only 60% to 75% of venture capital funds have survived more than 10 years. According to Harris, Jenkinson, and Kaplan (2013), since 2000 the average venture capital fund has underperformed public markets by about 5% over the life of the fund.

From 2001 to 2010, U.S. pension plans, on average, made 4.5% after fees from their investments in private equity funds. They paid an average of 4% each year in management fees, plus 20% of profits. This means fees amounted to about 70% of gross investment performance. Private equity funds usually require a five- to seven-year lockup period of invested capital. They have also been subject to widespread complaints of publishing inaccurate valuations.

Despite high fees, illiquidity, and variable performance, private equity is, by far, the Yale endowment’s largest asset allocation. Endowments, in general, are among the most prominent investors in private equity and hedge funds. Private equity funds, like hedge funds, are best left to institutional